Theorem 5.2

The equation of a circle with (x_1, y_1) and (x_2, y_2) as extremities of one of the diameters of the circle is $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$.

Proof

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the two extremities of the diameter AB, and $P(x, y)$ be any point on the circle. Then $\angle APB = \frac{\pi}{2}$ (angle in a semi-circle).

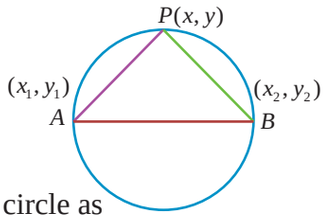


Fig.5.9

Therefore, the product of slopes of AP and PB is equal to -1.

That is, $\left(\frac{(y-y_1)}{(x-x_1)}\right)\left(\frac{(y-y_2)}{(x-x_2)}\right) = -1$ yielding the equation of the required circle as

$(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$. ■

Example 5.4

Find the general equation of the circle whose diameter is the line segment joining the points $(-4,-2)$ and $(1,1)$.

Solution

Equation of the circle with end points of the diameter as (x_1, y_1) and (x_2, y_2) given in theorem 5.2 is

$$(x-x_1)(x-x_2)+(y-y_1)(y-y_2) = 0$$

$$\Rightarrow \quad (x+4)(x-1)+(y+2)(y-1) = 0$$

$\Rightarrow \quad x^2+y^2+3x+y-6=0$ which is the required equation of the circle. ■

Theorem 5.3

The position of a point $P(x_1, y_1)$ with respect to a given circle $x^2+y^2+2gx+2fy+c=0$ in the plane containing the circle is outside or on or inside the circle according as

$$x_1^2+y_1^2+2gx_1+2fy_1+c \text{ is } \begin{cases} >0 & \text{or,} \\ =0 & \text{or,} \\ <0. \end{cases}$$

Proof

Equation of the circle is $x^2+y^2+2gx+2fy+c=0$ with centre C $(-g,-f)$ and radius $r=\sqrt{g^2+f^2-c}$.

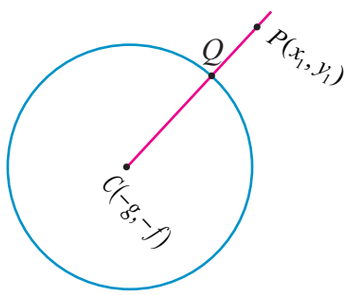


Fig.5.10

Let $P(x_1, y_1)$ be a point in the plane. Join CP and let it meet the circle at Q. Then the point P is outside, on or within the circle according as

$$|CP| \text{ is } \begin{cases} >|CQ| & \text{or,} \\ =|CQ| & \text{or,} \\ <|CQ|. \end{cases}$$

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